Abstract

Physicians and surgeons work in uncomfortable, awkward positions while performing strenuous, long, repetitive tasks causing them to be plagued by musculoskeletal pain and injuries during work-related operations. We aimed to focus on an array of different physicians and surgeons, assessing and identifying the prevalence of work-related MSDs injuries, the location and type of work-related MSDs, and the pain/injury/time-off from work percentage.

Methods: A literature review of published studies was conducted. The search was performed in PubMed and Google Scholar databases. The search terms were “musculoskeletal injuries among surgeons”, “ musculoskeletal injuries surgeons shoulder”, “musculoskeletal injuries med field”, “surgeon musculoskeletal injuries”, “MSK injuries of surgeons”, “injuries among neurosurgeons”, “MSK injuries via survey”, “reducing musculoskeletal disorders issues”, “WRMD among surgeons”, “musculoskeletal pain among surgeons”, and “injuries for surgeons”. Distributions of study characteristics were presented as summary statistics and percentages.

Results: The included articles were published between 1994 and 2025. The 27 studies comprised 5,531 participants in total (median 117, IQR 70-327, range 6-6000). Of the reported 25 out of 27 studies, the body regions most commonly represented were the neck (92.0%), shoulder (72.0%), wrist/hand (56.0%), back (44.0%), lower back (36.0%), elbow (36.0%), upper back (32.0%), leg/thigh (32.0%), knee (24.0%), ankle/foot (20.0%), arm/forearm (16%), hip (12.0%), and head (8.0%). Additionally, of only 8 of 27 studies, the specific diagnoses were recorded including carpal tunnel (62.5%), disc herniation (37.5%), rotator cuff/impinge (37.5%), epicondylitis (37.5% studies), neuropathy/numbness (37.5% studies), muscle spasm/fatigue (37.5% studies), tendonitis (25% studies), arthritis/OA (25% studies), headache (25% studies). Work-related musculoskeletal injuries resulted in 6.4% to 37.5% of respondents staying at home from work.

Conclusion: This literature review explored and uncovered the hidden dangers and high prevalence of work-related musculoskeletal disorders and injuries among surgeons and

physicians. Protecting the health and career longevity of health professionals is imperative for the wellness of all.

Keywords: Musculoskeletal injuries; Musculoskeletal pain for surgeons; Occupational injuries; Occupational diseases.